

Salvatore Barbagallo

Bioinformatician | Statistical Genetics · Transcriptomics · NGS

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PROFILE

Bioinformatician with an MSc in Bioinformatics and seven years in regulated clinical laboratories, including two in high-throughput clinical NGS. I build reproducible analyses in R, Python and Nextflow, most recently Mendelian randomisation, colocalisation and donor-aware single-nucleus RNA-seq for Parkinson's disease target prioritisation. From the lab side I bring a working knowledge of how sequencing data is generated, where it fails QC, and what audit-ready documentation requires.

BIOINFORMATICS PROJECTS

GNPMB and Parkinson's disease: Mendelian randomisation and colocalisation 2026

MSc dissertation · R · *TwoSampleMR* · *coloc* · *susieR* · *OpenGWAS* · *pQTL/eQTL* · [GitHub repository](#)

- Two-sample MR of circulating GPNMB on Parkinson's disease (ieu-b-7 GWAS) using SomaScan and UKB-PPP Olink pQTLs, with instrument QC ($F \geq 10$, $INFO \geq 0.8$), LD clumping at four thresholds, Wald ratio, fixed- and random-effects IVW and Cochran's Q.
- The SomaScan cis-instrument rs5850 gave OR 1.32 (95% CI 1.19–1.46; $P = 2.6 \times 10^{-7}$), and regional coloc.abf supported a shared signal ($PP.H4 = 0.94$ over 150 SNPs), tested for prior sensitivity and with SuSiE.
- Olink instruments were strongly heterogeneous and non-significant under random effects, so I concluded GPNMB warrants prioritisation rather than claiming causality. Released scripts, final instrument sets, sessionInfo and data provenance so every result can be reproduced.

GNPMB in Parkinson's substantia nigra: single-nucleus RNA-seq re-analysis 2026

Python · *Scanpy* · *PyDESeq2* · *R/speckle* · [GitHub repository](#)

- Re-analysed 83,484 nuclei from 29 donors (GSE243639; 15 PD, 14 controls); localised GPNMB mainly to microglia and re-clustered 12,995 microglial nuclei to characterise activation states.
- Treated the donor, not the nucleus, as the replicate: pseudobulk DE ($\sim$ age + sex + condition) found GPNMB about 2-fold higher in PD microglia ($\log_2FC$ 1.03, $P = 0.009$), which did not survive transcriptome-wide FDR (0.22). Tested composition with propeller and checked direction against GSE184950.

M. tuberculosis WGS drug-resistance workflow

Galaxy · *fastp* · *BWA-MEM2* · *Picard* · *bcftools* · *SnpEff* · *mosdepth* · *IGV* · [GitHub repository](#)

- Built an end-to-end workflow from SRA reads to annotated variants and ran it on 6 clinical isolates: 98–99.6% of reads mapped, with at least 52× depth across every target locus.
- Called rpoB (S450L, H445R, D435Y, L430P) and katG (S315T, Q295*) resistance mutations and confirmed them at read level in IGV. Classified 4 isolates as genotypically MDR-TB and 2 as rifampicin-resistant.

Containerised RNA-seq pipeline

Nextflow *DSL2* · *Docker* · *STAR* · *featureCounts* · *DESeq2* · [GitHub repository](#)

- Samplesheet-driven pipeline chaining FastQC, Cutadapt, STAR, featureCounts, MultiQC and DESeq2 in versioned containers. Validated end-to-end on bundled test data, with Nextflow trace, report and timeline outputs.

EXPERIENCE

Bioinformatician, Scientific AI Evaluation · Mercor (contract, remote) Jan 2026 – present

- Authored 106 accepted tasks (Jan–Jul 2026) for a frontier model's bioinformatics benchmark. Each is a hidden Python function implementing a cited genomics or biostatistics rule, with unit tests, that the model must reverse-engineer by choosing inputs and reading outputs.
- Calibrated difficulty from model trajectory data using ablation ladders and adversarial foils: textbook solutions scored below 50% while the full solution reached 100%. 80% first-pass approval in peer and client review.
- Currently: expert evaluation of AI agent runs on preclinical drug R&D tasks, and blinded A/B comparisons of coding models on large open-source codebases. Earlier AI evaluation contracts with Outlier and Micro1.

Specialist Biomedical Scientist, Stem Cell Laboratory · UCL Hospitals NHS Trust Sep 2021 – Dec 2025

- Built Python/Excel tracking tools to replace manual inventory reconciliation, reducing inventory errors by ~30% and saving ~10 hours a week.
- Managed traceability and audit-ready records for 11 clinical and ATMP trials, including CAR-T, and led the digitisation of SOPs and QA documentation.
- Processed and cryopreserved PBSC, bone marrow, DLI and CD34+ products, and generated CD3+/CD34+ flow cytometry data for time-critical clinical decisions.

Laboratory Scientist, Clinical Genomics · CooperGenomics, London Jul 2019 – Sep 2021

- Prepared and QC'd NGS libraries for PGT-A, PGT-SR and PGT-M (96–192 samples per run, 500+ samples a week) to clinical turnaround targets.
- Programmed and validated Mosquito HV and Dragonfly liquid handlers, and wrote and reviewed SOPs to ISO standards.

Biomedical Laboratory Assistant, Cytology · Leicester Royal Infirmary Dec 2018 – Jun 2019

EDUCATION

MSc Bioinformatics · Atlantic Technological University (remote) Sep 2025 – Sep 2026
Dissertation: Prioritising GPNMB as a Parkinson's disease-relevant protein using Mendelian randomisation and colocalisation

MSc Cell and Gene Therapy · University College London Sep 2021 – Sep 2023
Dissertation: Expansion and preservation of haematopoietic potential in human amniotic fluid stem cells

BSc Biomedical Science · University of Catania, Italy Sep 2014 – Oct 2017

TECHNICAL SKILLS

Programming: Python (pandas, Scanpy, PyDESeq2, scikit-learn), R (TwoSampleMR, coloc, susieR, DESeq2, tidyverse), Bash, SQL

Workflows: Nextflow (DSL2), Docker, Galaxy, Conda, Git, Linux

NGS: FastQC, fastp, Cutadapt, STAR, featureCounts, BWA-MEM2, samtools, Picard, bcftools, SnpEff, mosdepth, MultiQC, IGV

Statistical genetics: two-sample MR, Bayesian colocalisation, SuSiE fine-mapping, LD clumping, GWAS / pQTL / eQTL summary statistics

Single-cell: snRNA-seq QC and clustering, donor-level pseudobulk DE, cell-type composition testing

Training: Johns Hopkins Genomic Data Science Specialization; Wellcome Bioinformatics for Biologists; IBM Machine Learning

Languages: Italian (native), English (fluent), Portuguese (fluent), Spanish (advanced)